

Nithan B S

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Summary

Data Science and Machine Learning professional specializing in NLP, deep learning architectures, and GenAI workflows. Hands-on experience building, optimizing, containerizing, and deploying functional applications using RAG frameworks, transformers, Docker, and automated APIs.

Technical Skills

- **Data Engineering:** ETL Pipelines, MySQL, Database Design, API Integration, Git Version Control.
- **Data Science & ML:** Python (Scikit-learn, Pandas), SQL, Statistical Modelling.
- **AI & Deep Learning:** GenAI, LangChain, RAG Pipelines, BERT, NLP, Neural Networks, Hugging Face Transformers.
- **Deployment & Tools:** Streamlit, FastAPI, Docker, AWS, Hugging Face Spaces.

Key Projects

Legal Intelligence Assistant - Retrieval-Augmented Generation (RAG) Pipeline | [Live Demo](#)

- Built a rapid, high-precision retrieval of specific clauses from extensive legal document corpora.
- Implemented an end-to-end RAG system using vector embeddings and semantic search mechanisms. Developed custom evaluation workflows incorporating hallucination-aware metrics to ensure output reliability.
- Successfully deployed a RAG system achieving an **average RAGAS score of 0.77**, with high retrieval precision demonstrated by a **faithfulness score of >0.80**, **answer relevance of 0.70**, and **context relevance of 0.77-0.79**.
- *Tools and Techniques:* Python, LangChain, ChromaDB, Docker, Hugging Face, Git.

Financial News Sentiment Prediction using Deep Learning & BERT | [Live Demo](#)

- Established a system to classify financial sentiment, requiring robust performance against standard textual datasets.
- Developed deep learning models (RNN, LSTM, GRU) from scratch utilizing personalized training loops. Benchmarked this architecture against state-of-the-art FinBERT models using Hugging Face Transformers.
- Achieved a **macro F1-score of 0.72 with a GRU** architecture, while FinBERT yielded a superior **0.84 macro F1-score** with clear efficacy in identifying minority bullish and bearish signals.
- *Tools and Techniques:* Python, BERT, PyTorch, Docker, Hugging Face Hub.

Marketing Campaign Analysis & Text-to-SQL Agent | [Live Demo](#)

- Solved data inconsistencies and created a mechanism for natural language retrieval of structured data.
- Executed comprehensive Exploratory Data Analysis (EDA) and data cleaning. Engineered custom customer segmentation logic and a GenAI layer using a 70B Llama model for automated chart generation.
- Created an interactive dashboard displaying customer behaviour pattern with natural language querying capabilities.
- *Tools and Techniques:* Python, SQL, Llama 70B, Groq API, Streamlit.

Professional Development

Independent Researcher & Competitive Examination Preparation | Chennai | 2020 – 2025

- Pursued full-time preparation for highly competitive national-level examinations, strengthening analytical reasoning, research, problem-solving, and structured decision-making skills.
- Qualified UGC-NET (2021) and PhD Entrance Examinations in Environmental Sciences.
- Conducted independent study across public policy, economics, governance, environment, and data interpretation.

Intern | Regional Integrated Multi-Hazard Early Warning System (RIMES) | 2019

- Assisted in hazard modelling, geospatial analysis, and environmental data processing for disaster risk reduction initiatives.
- Supported data-driven assessments for early warning systems and climate resilience planning.
- Collaborated with researchers on analytical workflows involving environmental and hazard datasets.

Education

- Master Data Science | GUVI | 2026
- Integrated M.Sc., Environmental Sciences | Central University of Rajasthan | 2015-2020

Certificates

- **Generative AI Workshop** | Understanding RAG, AI Agents and Agentic AI | GUVI | 2026
- **Python - IITM Pravartak Certification** | GUVI | 2026
- **Paper Presentation** | International workshop on Extreme Severe Storms and Disaster Mitigation Strategy – 2020 (ESSDMS2) | Sponsored by DPRI, Japan | 2020.